**Code No: 133AJ** **R16**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech II Year I Semester Examinations, April/May - 2023**
**DIGITAL LOGIC DESIGN**
**(Common to CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

**PART – A**

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | What is meant by self-complementary code? Give examples. | [2] |
| b) | Implement the basic gates using only NAND gates. | [3] |
| c) | What are the disadvantages of Karnaugh map? | [2] |
| d) | Simplify the following Boolean function to two literals. $ABC'D + A'BD + ABCD$ | [3] |
| e) | Implement 2-input NAND gate using 2-to-1 multiplexer. | [2] |
| f) | Draw the 4 to 2 encoder logic diagram. | [3] |
| g) | Differentiate a latch and a flip-flop. | [2] |
| h) | Draw the excitation table for JK flip-flop. | [3] |
| i) | Define RTL and which one is the basic gate in this logic. | [2] |
| j) | Differentiate simple PLDs and Sequential PLDs. | [3] |

**PART – B**

**(50 Marks)**

2.a) What is the importance of representing a number in compliment form? Explain different signed magnitude representations of binary numbers.
b) Discuss the properties and applications of XOR gates. [6+4]

**OR**

3. Minimize the following Boolean expression using algebraic manipulation and draw its logic diagram.
$f(A,B,C,D) = ABC'D' + ABC'D + AB'C'D + ABCD + AB'CD + ABCD' + AB'CD'$ [10]

4. A warning busser is to sound when the following conditions apply:
a) Switches A, B, C are on.
b) Switches A and B are on but switch C is off.
c) Switches A and C are on but switch B is off.
d) Switches B and C are on but switch A is off.
Draw a truth table and obtain the simplified Boolean expression using K-map. [10]

**OR**

5. Implement a 4-bit even parity checker using 8 to 1 multiplexer. [10]

JNTUH Used papers 2023

6.a) Design a 4-bit magnitude comparator.
b) Design a 16 × 1 multiplexer using 4 × 1 multiplexers and draw its diagram. [5+5]

**OR**

7.a) Design a code converter that converts a decimal digit from 8,4,–2,–1 code to BCD.
b) Draw the 4-bit parallel adder circuit diagram. [6+4]

8. Design a 3-bit asynchronous binary up counter using T flip-flops. [10]

**OR**

9. Show that a Johnson counter with *n* flip-flops produces a sequence of 2 *n* states. List the 10 states produced with five flip-flops and explain with a neat logic diagram. [10]

10. Implement 2-bit squarer circuit using programmable array logic. [10]

**OR**

11. Explain the arithmetic and logical micro operations with examples. [10]

**---ooOoo---**

JNTUH Used papers 2023